

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): HD10K-M, DS+6K-M, HD6K-M, WU7K-M, WX10K-M, DS+10K-M, WU12K-M, DS+14K-M, HD14K-M, WU14K-M

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0002-b002	1.4.2

Version History

Module Version	Date	Notes
1_0_3_0	9/5/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='HD10K-M')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='HD10K-M')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3002, Model='HD10K-M')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command 3DInvert	Value 'On'	Value 'Off'	
# 3DInvert example InterfaceName.Set('3DInvert', 'On')			
Command 3DMode	Value 'Off'	Value 'Auto'	Value 'Native'
	'Frame Doubled'	'Dual Input'	'Side-by-Side'
	'Top/Bottom'	'Frame Packed'	
# 3DMode example InterfaceName.Set('3DMode', 'Off')			
Command 3DSync	Value 'On'	Value 'Off'	
# 3DSync example InterfaceName.Set('3DSync', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Brightness	Value -1000 to 1000 in steps of 1		
# Brightness example InterfaceName.Set('Brightness', 1000)			
Command Channel	Value '1' – '99'		
# Channel example InterfaceName.Set('Channel', '1')			
Command Contrast	Value 0 to 1000 in steps of 1		
# Contrast example InterfaceName.Set('Contrast', 1000)			
Command Focus	Value 'In'	Value 'Out'	Value 'Stop'
# Focus example InterfaceName.Set('Focus', 'In')			

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Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command HorizontalShift	Value 'Left'	Value 'Right'	Value 'Stop'
# HorizontalShift example InterfaceName.Set('HorizontalShift', 'Left')			
Command Input	Value 'SLOT 1-1' 'SLOT 1-4' 'SLOT 2-1' 'SLOT 2-4' 'SLOT 3-1' 'SLOT 3-4' 'SLOT 4-1' 'SLOT 4-4'	Value 'SLOT 1-2' 'SLOT 1-5' 'SLOT 2-2' 'SLOT 2-5' 'SLOT 3-2' 'SLOT 3-5' 'SLOT 4-2' 'SLOT 4-5'	Value 'SLOT 1-3' 'SLOT 1-6' 'SLOT 2-3' 'SLOT 2-6' 'SLOT 3-3' 'SLOT 3-6' 'SLOT 4-3' 'SLOT 4-6'
# Input example InterfaceName.Set('Input', 'SLOT 1-1')			
Command LampOperation	Value 'Auto-select' 'Dual Lamp'	Value 'Lamp 1'	Value 'Lamp 2'
# LampOperation example InterfaceName.Set('LampOperation', 'Auto-select')			
Command PictureInPicture	Value 'Disabled'	Value 'Enabled'	Value 'Picture-by-Picture'
# PictureInPicture example InterfaceName.Set('PictureInPicture', 'Disabled')			
Command PictureInPicturePosition	Value 'Top Right' 'Bottom Right'	Value 'Top Left'	Value 'Bottom Left'
# PictureInPicturePosition example InterfaceName.Set('PictureInPicturePosition', 'Top Right')			
Command PictureInPictureSwap	Value None		
# PictureInPictureSwap example InterfaceName.Set('PictureInPictureSwap', None)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

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Command Shutter	Value 'Open'	Value 'Close'	
# Shutter example InterfaceName.Set('Shutter', 'Open')			
Command SplitScreen	Value 'Off'	Value 'Side'	Value 'Top'
# SplitScreen example InterfaceName.Set('SplitScreen', 'Off')			
Command VerticalShift	Value 'Up'	Value 'Down'	Value 'Stop'
# VerticalShift example InterfaceName.Set('VerticalShift', 'Up')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Zoom	Value 'In'	Value 'Out'	Value 'Stop'
# Zoom example InterfaceName.Set('Zoom', 'In')			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command 3DInvert	Value 'On'	Value 'Off'	
# 3DInvert examples InterfaceName.Update('3DInvert') Value = InterfaceName.ReadStatus('3DInvert') InterfaceName.SubscribeStatus('3DInvert', None, FeedbackHandler)			
Command 3DMode	Value 'Off'	Value 'Auto'	Value 'Native'
	'Frame Doubled'	'Dual Input'	'Side-by-Side'
	'Top/Bottom'	'Frame Packed'	
# 3DMode examples InterfaceName.Update('3DMode') Value = InterfaceName.ReadStatus('3DMode') InterfaceName.SubscribeStatus('3DMode', None, FeedbackHandler)			
Command 3DSync	Value 'On'	Value 'Off'	
# 3DSync examples InterfaceName.Update('3DSync') Value = InterfaceName.ReadStatus('3DSync') InterfaceName.SubscribeStatus('3DSync', None, FeedbackHandler)			
Command Brightness	Value -1000 to 1000 in steps of 1		
# Brightness examples InterfaceName.Update('Brightness') Value = InterfaceName.ReadStatus('Brightness') InterfaceName.SubscribeStatus('Brightness', None, FeedbackHandler)			
Command Channel	Value '1' – '99'		
# Channel examples			

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InterfaceName.Update('Channel') Value = InterfaceName.ReadStatus('Channel') InterfaceName.SubscribeStatus('Channel', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value		
Contrast	0 to 1000 in steps of 1		
# Contrast examples InterfaceName.Update('Contrast') Value = InterfaceName.ReadStatus('Contrast') InterfaceName.SubscribeStatus('Contrast', None, FeedbackHandler)			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'SLOT 1-1'	'SLOT 1-2'	'SLOT 1-3'
	'SLOT 1-4'	'SLOT 1-5'	'SLOT 1-6'
	'SLOT 2-1'	'SLOT 2-2'	'SLOT 2-3'
	'SLOT 2-4'	'SLOT 2-5'	'SLOT 2-6'
	'SLOT 3-1'	'SLOT 3-2'	'SLOT 3-3'
	'SLOT 3-4'	'SLOT 3-5'	'SLOT 3-6'
	'SLOT 4-1'	'SLOT 4-2'	'SLOT 4-3'
	'SLOT 4-4'	'SLOT 4-5'	'SLOT 4-6'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	Value
LampOperation	'Auto-select'	'Lamp 1'	'Lamp 2'
	'Dual Lamp'		
# LampOperation examples InterfaceName.Update('LampOperation') Value = InterfaceName.ReadStatus('LampOperation') InterfaceName.SubscribeStatus('LampOperation', None, FeedbackHandler)			
Command	Value		
LampUsage ¹	Hours		
Qualifier Key	Qualifier Value		

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'Lamp'	'1' – '2'		
# LampUsage examples InterfaceName.Update('LampUsage', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command LampUsage	Value 0 –		
Qualifier Key 'Lamp'	Qualifier Value '1' – '2'		
# LampUsage examples InterfaceName.Update('LampUsage', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command PictureInPicture	Value 'Disabled'	Value 'Enabled'	Value 'Picture-by-Picture'
# PictureInPicture examples InterfaceName.Update('PictureInPicture') Value = InterfaceName.ReadStatus('PictureInPicture') InterfaceName.SubscribeStatus('PictureInPicture', None, FeedbackHandler)			
Command PictureInPicturePosition	Value 'Top Right' 'Bottom Right'	Value 'Top Left'	Value 'Bottom Left'
# PictureInPicturePosition examples InterfaceName.Update('PictureInPicturePosition') Value = InterfaceName.ReadStatus('PictureInPicturePosition') InterfaceName.SubscribeStatus('PictureInPicturePosition', None, FeedbackHandler)			
Command Power	Value 'On' 'Cooling'	Value 'Off'	Value 'Warming'
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command Shutter	Value 'Open'	Value 'Close'	
# Shutter examples InterfaceName.Update('Shutter') Value = InterfaceName.ReadStatus('Shutter') InterfaceName.SubscribeStatus('Shutter', None, FeedbackHandler)			
Command SplitScreen	Value 'Off'	Value 'Side'	Value 'Top'
# SplitScreen examples InterfaceName.Update('SplitScreen') Value = InterfaceName.ReadStatus('SplitScreen') InterfaceName.SubscribeStatus('SplitScreen', None, FeedbackHandler)			

Command	Value	Value
VideoMute	'On'	'Off'
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>		

¹ Lamp Usage requires 'Lamp' to be set.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

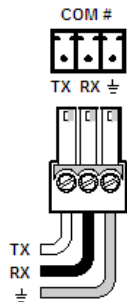
Data Bits: 8

Parity: None

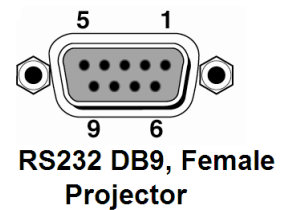
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection	No
Capabilities:	
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

3D Invert Off	(TDN0)
3D Invert On	(TDN1)
3D Mode Auto	(TDM1)
3D Mode Dual Input	(TDM4)
3D Mode Frame Doubled	(TDM3)
3D Mode Frame Packed	(TDM7)
3D Mode Native	(TDM2)
3D Mode Off	(TDM0)
3D Mode Side-by-Side	(TDM5)
3D Mode Top/Bottom	(TDM6)
3D Sync Off	(TD00)
3D Sync On	(TD01)
Auto Image None	(ASU)
Brightness 1000	(BRT1000)
Brightness -1000	(BRT-1000)
Channel 1	(CHA1)
Channel 99	(CHA99)
Contrast 0	(CON0)
Contrast 1000	(CON1000)
Focus In	(LMV+FRUN1)
Focus Out	(LMV+FRUN-1)
Focus Stop	(LMV+FRUN0)
Freeze Off	(FRZ0)
Freeze On	(FRZ1)
Horizontal Shift Left	(LMV+HRUN1)
Horizontal Shift Right	(LMV+HRUN-1)
Horizontal Shift Stop	(LMV+HRUN0)
Input SLOT 1-1	(SIN11)
Input SLOT 1-2	(SIN12)
Input SLOT 1-3	(SIN13)
Input SLOT 1-4	(SIN14)
Input SLOT 1-5	(SIN15)
Input SLOT 1-6	(SIN16)
Input SLOT 2-1	(SIN21)
Input SLOT 2-2	(SIN22)
Input SLOT 2-3	(SIN23)

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Input SLOT 2-4	(SIN24)
Input SLOT 2-5	(SIN25)
Input SLOT 2-6	(SIN26)
Input SLOT 3-1	(SIN31)
Input SLOT 3-2	(SIN32)
Input SLOT 3-3	(SIN33)
Input SLOT 3-4	(SIN34)
Input SLOT 3-5	(SIN35)
Input SLOT 3-6	(SIN36)
Input SLOT 4-1	(SIN41)
Input SLOT 4-2	(SIN42)
Input SLOT 4-3	(SIN43)
Input SLOT 4-4	(SIN44)
Input SLOT 4-5	(SIN45)
Input SLOT 4-6	(SIN46)
Lamp Operation Auto-select	(LOP0)
Lamp Operation Dual Lamp	(LOP3)
Lamp Operation Lamp 1	(LOP1)
Lamp Operation Lamp 2	(LOP2)
Picture In Picture Disabled	(PIP0)
Picture In Picture Enabled	(PIP1)
Picture In Picture Picture-by-Picture	(PIP2)
Picture In Picture Position Bottom Left	(PPP2)
Picture In Picture Position Bottom Right	(PPP3)
Picture In Picture Position Top Left	(PPP1)
Picture In Picture Position Top Right	(PPP0)
Picture In Picture Swap None	(PPS)
Power Off	(PWR0)
Power On	(PWR1)
Shutter Close	(SHU 1)
Shutter Open	(SHU 0)
Split Screen Off	(SPT0)
Split Screen Side	(SPT1)
Split Screen Top	(SPT2)
Vertical Shift Down	(LMV+VRUN-1)
Vertical Shift Stop	(LMV+VRUN0)
Vertical Shift Up	(LMV+VRUN1)
Video Mute Off	(PMT0)
Video Mute On	(PMT1)

**Global Scripter Module
Communication Sheet**

Zoom In	(LMV+ZRUN1)
Zoom Out	(LMV+ZRUN-1)
Zoom Stop	(LMV+ZRUN0)

Appendix B. Update Commands

3D Invert	(TDN?)
3D Mode	(TDM?)
3D Sync	(TDO?)
Brightness	(BRT?)
Channel	(CHA?)
Contrast	(CON?)
Freeze	(FRZ?)
Input	(SIN?)
Lamp Operation	(LOP?)
Lamp Usage Lamp 1	(HIS+LMP1?)
Lamp Usage Lamp 2	(HIS+LMP2?)
Picture In Picture	(PIP?)
Picture In Picture Position	(PPP?)
Power	(PWR?)
Shutter	(SHU?)
Split Screen	(SPT?)
Video Mute	(PMT?)